

Indian Statistical Institute

BSDS - Second Year

Exercise Set 1.

1. Let P, P_1 and P_2 be pgfs.
 - a) Show that $\lambda P_1 + (1 - \lambda)P_2$ is a pgf for any $0 \leq \lambda \leq 1$. Show that $P_1 P_2$ is a pgf.
 - b) Is $\exp(P(s) - 1)$ a pgf? Fix $0 < \lambda < 1$. Is $P(\lambda s)/P(\lambda)$ necessarily a pgf?
2. Find the pgf for the following random variables and state their radius of convergence:
 - a) X where $\mathbb{P}(X = n) = \frac{c}{n!}$ for $n \geq 0$. Find the constant c first.
 - b) Y where $\mathbb{P}(Y = n) = \frac{c}{(n+1)(n+2)}$ for $n \geq 0$. Find the constant c first.
 - c) Z where $\mathbb{P}(Z = n) = \frac{1}{2^{n+1}}$ for $n \geq 0$.
3. Find the distribution of the random variables in each of the cases and find $\mathbb{P}(X \geq n)$ for $n \geq 1$.
 - a) $P(s) = \frac{1}{3-2s}$;
 - b) $P(s) = \frac{1}{4-3s}$.
 - c) Two dice are rolled. Let X be the sum of the outcomes. Find the pgf of X . Use this pgf to calculate the probability that the sum is divisible by 3.
 - d) If the pgf of a random variable X is $P(s) = \frac{1}{2}e^s + \frac{1}{4}s^2 + \frac{1}{4}$, find $\mathbb{P}(X \geq 2)$.
4.
 - a) Use generating functions to prove the combinatorial identity: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.
 - b) Find the sequence whose generating function is $\frac{1}{(1-s)^4}$ by expressing it as a product of known generating functions and using convolution.
 - c) Let a_n be the number of ways to tile a $2 \times n$ board with 1×2 dominoes for $n \geq 1$. Find a recurrence relation for a_n for $n \geq 3$. Set a_1, a_2 appropriately and, find its generating function.

5. a) Let p_k be the probability that three consecutive Heads (HHH) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- b) Let p_k be the probability of the pattern of two consecutive Heads (HH) or the pattern of Head followed by Tail (HT) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- c) Let p_k be the probability that the total obtained in k throws of a dice (not necessarily fair) is divisible by 3. Find the generating function of $\{p_k : k \geq 0\}$. Find $\lim_{k \rightarrow \infty} p_k$.
6. a) Claims to an insurance company arrive as a Poisson process with rate λ . The size of each claim is an **iid** random variable Y with pgf $P_Y(s)$. Let S_n be the total claim amount up to time n . Find the pgf of S_n .
- b) A hen lays N eggs, where $N \sim \text{Poisson}(\lambda)$. Each egg hatches with probability p independently. Let K be the number of chicks. Find the pgf of K and identify its distribution.
- c) In the previous problem, find the expected number of chicks, $\mathbb{E}[K]$, using the random sum formula and verify by direct computation.